



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MV & SONS PETROLEUM

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

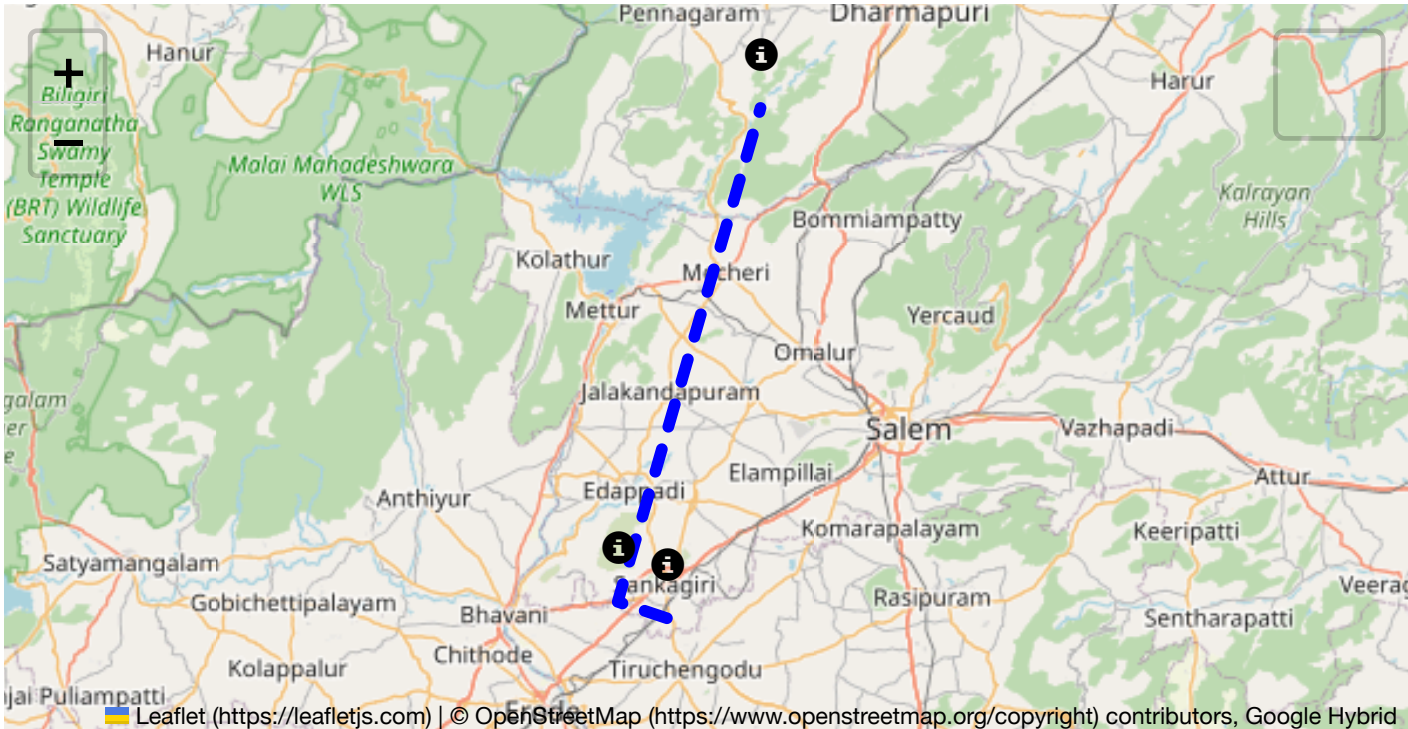
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

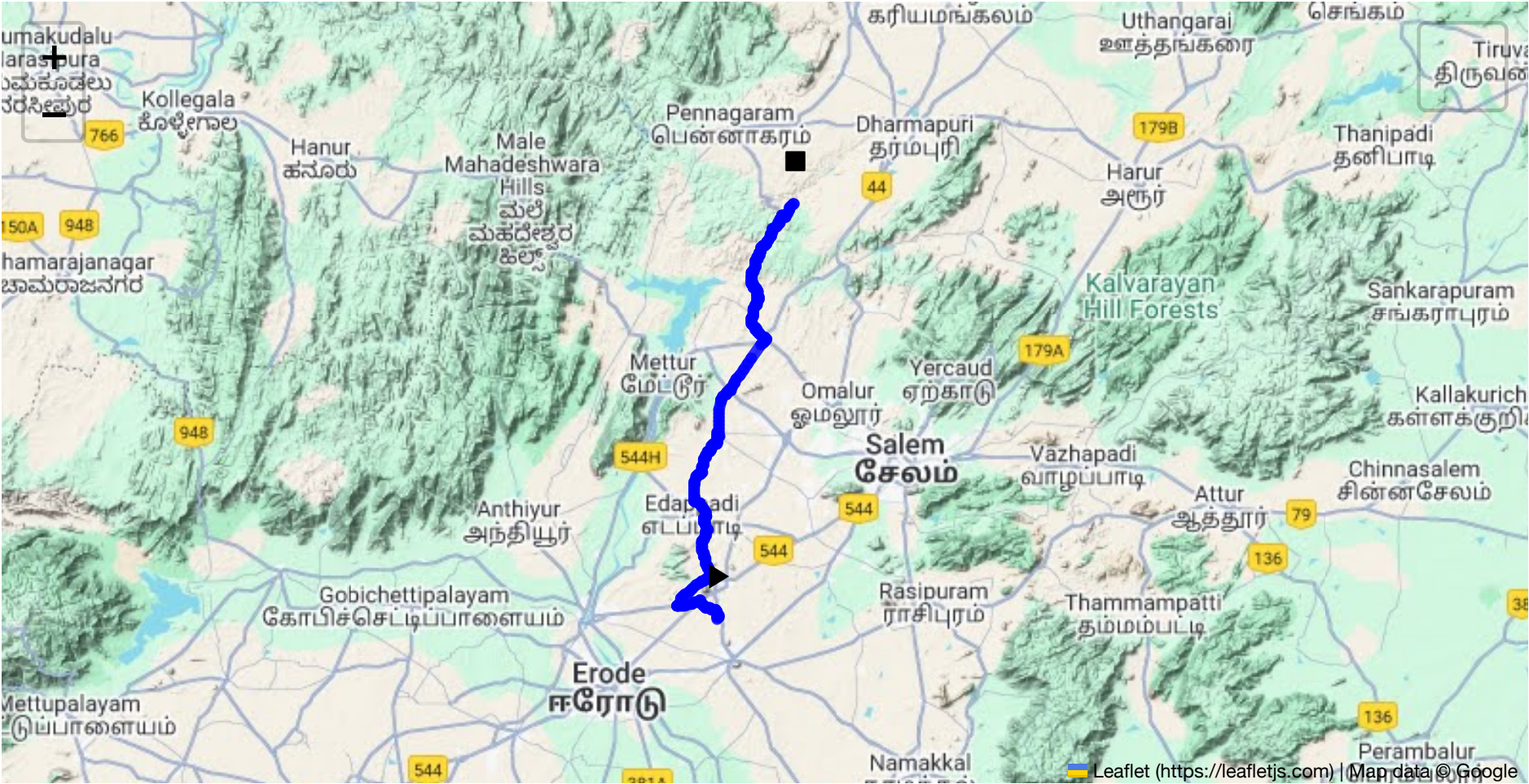
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 87.14 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours

Start: (11.4381, 77.8734)
End: (12.02326, 77.984317)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W in Sangagiri to SF No 407/1A in Chinnampalli spans approximately 87.14 kilometers. The journey typically takes around 2.15 hours for heavy vehicles carrying hazardous materials. The route passes through areas such as Padaiveedu, following the Bhavani Main Road.

Weather Conditions

Typical weather conditions in Tamil Nadu include a hot and humid climate with distinct monsoon seasons. The region experiences heavy rainfall during the Northeast Monsoon (October to December). Potential hazards include wet and slippery roads, reduced visibility, and flooding, particularly during the monsoon.

Traffic Patterns

The route may experience moderate to heavy traffic in the morning (8 AM - 10 AM) and evening (5 PM - 8 PM) due to local commute. Areas near towns such as Sangagiri and Bhavani tend to have congestion, especially around major intersections and market areas.

Road Quality and Infrastructure

Roads along this route vary in quality. Highways are generally well-maintained, but rural roads may have potholes and uneven surfaces. Watch for narrow stretches and sharp turns, especially near Padaiveedu. Infrastructure limitations include the absence of shoulders and street lighting in some areas.

Alternative Routes

In case of emergencies, consider the State Highway 6 (SH 6) as an alternative. This route may increase travel time but provides access to larger urban centers with better facilities.

Regulations on Hazardous Material Transport

Local regulations require permits for transporting hazardous materials within Tamil Nadu. Restrictions apply, particularly in densely populated areas, to minimize risk to the public. Ensure compliance with safety standards and legal requirements.

Historical Incidents

While specific data on historical incidents involving heavy vehicles on this route is scarce, incidents often include vehicle rollovers and collisions due to narrow roads and sharp curves. Stay vigilant, particularly during adverse weather.

Environmental Considerations

Be aware of agricultural zones and proximity to water bodies, which are sensitive to contamination. Reduce speed and follow designated routes to minimize environmental impact.

Communication Coverage

Cellular coverage varies; urban areas generally have good network access, but rural stretches can have dead zones. Use GPS devices with offline capabilities and prepare for periods without communication.

Emergency Response Times

Emergency response times can vary. Urban areas like Sangagiri and Bhavani can expect quicker responses (30-45 mins), while rural segments near Padaiveedu may experience delays (up to 90 mins). Familiarize yourself with local emergency numbers and maintain regular check-ins.

Overall Risk Assessment

This route poses moderate risks due to weather, traffic, and road conditions, particularly when transporting hazardous materials. Mitigation measures include scheduling travel outside peak hours, avoiding travel during heavy rain, and ensuring compliance with transport regulations. Prepare for communication challenges and longer emergency response times in rural areas.

By taking adequate precautions, training drivers thoroughly, and adopting flexible routing, safety along this route can be significantly improved.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45781, 77.85203	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45636, 77.81678	15 KM/Hr
10	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
11	Turn	Medium	11.49441, 77.86690	30 KM/Hr
12	Turn	Medium	11.49971, 77.86468	30 KM/Hr
13	Turn	Medium	11.50885, 77.86037	30 KM/Hr
14	Turn	Medium	11.52519, 77.85535	30 KM/Hr
15	Turn	Medium	11.54790, 77.85296	30 KM/Hr
16	Turn	Medium	11.55499, 77.85656	30 KM/Hr
17	Turn	High	11.57893, 77.85430	15 KM/Hr
18	Turn	Medium	11.57919, 77.85585	30 KM/Hr
19	Turn	Medium	11.58037, 77.85749	30 KM/Hr
20	Turn	Medium	11.60978, 77.84165	30 KM/Hr
21	Turn	Medium	11.61795, 77.84020	30 KM/Hr
22	Turn	Medium	11.63085, 77.84072	30 KM/Hr
23	Turn	Medium	11.64073, 77.84627	30 KM/Hr
24	Turn	Medium	11.64130, 77.84816	30 KM/Hr
25	Turn	Medium	11.64547, 77.85129	30 KM/Hr
26	Turn	Medium	11.65274, 77.85197	30 KM/Hr
27	Turn	Medium	11.66653, 77.85622	30 KM/Hr
28	Turn	High	11.67310, 77.86031	15 KM/Hr
29	Turn	Medium	11.67357, 77.86193	30 KM/Hr
30	Turn	Medium	11.68196, 77.86690	30 KM/Hr
31	Turn	High	11.68492, 77.86898	15 KM/Hr
32	Turn	High	11.68463, 77.87262	15 KM/Hr
33	Turn	High	11.69455, 77.87583	15 KM/Hr
34	Turn	High	11.69558, 77.87439	15 KM/Hr
35	Turn	High	11.76019, 77.89047	15 KM/Hr
36	Turn	High	11.75964, 77.89200	15 KM/Hr
37	Blind Spot	Blind Spot	11.83264, 77.94408	10 KM/Hr
38	Turn	Medium	11.84317, 77.93283	30 KM/Hr
39	Turn	Medium	11.84593, 77.92925	30 KM/Hr
40	Turn	Medium	11.85102, 77.92587	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
41	Turn	Medium	11.88158, 77.93124	30 KM/Hr
42	Turn	Medium	11.88495, 77.93258	30 KM/Hr
43	Turn	Medium	11.88951, 77.93278	30 KM/Hr
44	Turn	High	11.88979, 77.93330	15 KM/Hr
45	Turn	Medium	11.89093, 77.93364	30 KM/Hr
46	Turn	Medium	11.89304, 77.93241	30 KM/Hr
47	Turn	High	11.89818, 77.93069	15 KM/Hr
48	Turn	Medium	11.89893, 77.93090	30 KM/Hr
49	Turn	Medium	11.89967, 77.93212	30 KM/Hr
50	Turn	Medium	11.90023, 77.93247	30 KM/Hr
51	Turn	Medium	11.90083, 77.93245	30 KM/Hr
52	Turn	Medium	11.90434, 77.93007	30 KM/Hr
53	Turn	High	11.90535, 77.92665	15 KM/Hr
54	Turn	Medium	11.90979, 77.92570	30 KM/Hr
55	Turn	High	11.91187, 77.92304	15 KM/Hr
56	Turn	High	11.91332, 77.92336	15 KM/Hr
57	Turn	High	11.91408, 77.92286	15 KM/Hr
58	Turn	Medium	11.91463, 77.92308	30 KM/Hr
59	Blind Spot	Blind Spot	11.92989, 77.92663	10 KM/Hr
60	Blind Spot	Blind Spot	11.93022, 77.92591	10 KM/Hr
61	Turn	Medium	11.94153, 77.92988	30 KM/Hr
62	Turn	Medium	11.94365, 77.92965	30 KM/Hr
63	Turn	Medium	11.94447, 77.93046	30 KM/Hr
64	Turn	Medium	11.94615, 77.93045	30 KM/Hr
65	Turn	Medium	11.94962, 77.93394	30 KM/Hr
66	Turn	High	11.95109, 77.93458	15 KM/Hr
67	Turn	Medium	11.95167, 77.93416	30 KM/Hr
68	Blind Spot	Blind Spot	11.95462, 77.93515	10 KM/Hr
69	Turn	High	11.95442, 77.93544	15 KM/Hr
70	Blind Spot	Blind Spot	11.95363, 77.93517	10 KM/Hr
71	Turn	Medium	11.95431, 77.93560	30 KM/Hr
72	Turn	Medium	11.95642, 77.93570	30 KM/Hr
73	Turn	High	11.96346, 77.93881	15 KM/Hr
74	Turn	Medium	11.96404, 77.94036	30 KM/Hr
75	Turn	Medium	11.97772, 77.95193	30 KM/Hr
76	Turn	Medium	11.97993, 77.95175	30 KM/Hr
77	Turn	Medium	11.99222, 77.96011	30 KM/Hr
78	Turn	Medium	11.99874, 77.96519	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
79	Blind Spot	Blind Spot	12.00648, 77.96642	10 KM/Hr
80	Turn	High	12.00377, 77.97059	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
8	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
9	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
10	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
11	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
12	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
13	hospital	Rishan Hospital	11.879358, 77.9286133	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49441, 77.86690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49971, 77.86468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50885, 77.86037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52519, 77.85535



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54790, 77.85296



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55499, 77.85656



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57919, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58037, 77.85749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60978, 77.84165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63085, 77.84072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64073, 77.84627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64130, 77.84816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64547, 77.85129



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65274, 77.85197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66653, 77.85622



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67310, 77.86031



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67357, 77.86193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68492, 77.86898



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69455, 77.87583



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69558, 77.87439



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75964, 77.89200



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.83264, 77.94408



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84317, 77.93283



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84593, 77.92925



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.85102, 77.92587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88158, 77.93124



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88495, 77.93258



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88951, 77.93278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.88979, 77.93330



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89093, 77.93364



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89304, 77.93241



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.89818, 77.93069



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89893, 77.93090



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89967, 77.93212



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90023, 77.93247



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90083, 77.93245



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90434, 77.93007



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.90535, 77.92665



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.90979, 77.92570



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91187, 77.92304



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91332, 77.92336



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.91408, 77.92286



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91463, 77.92308



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.92989, 77.92663



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.93022, 77.92591



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94153, 77.92988



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94365, 77.92965



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94447, 77.93046



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94615, 77.93045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94962, 77.93394



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95109, 77.93458



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95167, 77.93416



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.95462, 77.93515



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95442, 77.93544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.95363, 77.93517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95431, 77.93560



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95642, 77.93570



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96346, 77.93881



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96404, 77.94036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97772, 77.95193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.97993, 77.95175



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99222, 77.96011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99874, 77.96519



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.00648, 77.96642



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.00377, 77.97059
